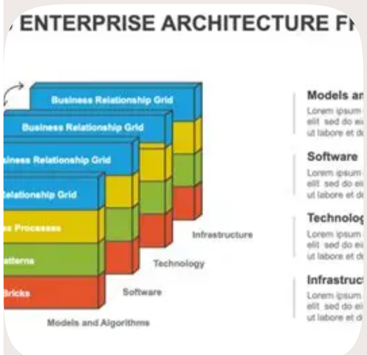
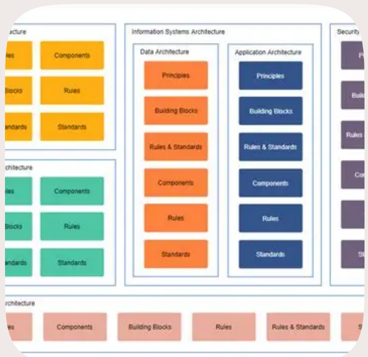
AS/400 → Snowflake Migration WBS (Work Breakdown Structure)

Level 1 → Level 4 decomposition

1. Program Initiation & Governance





1.1 Program Governance

- **Define governance model**
- Steering committee setup
- RACI matrix creation

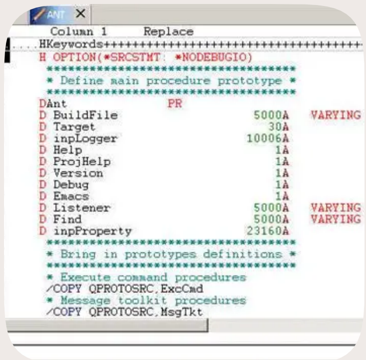
1.2 Architecture & Security

- **Snowflake environment readiness**
- IAM roles & RBAC/ABAC
- Network connectivity (VPN, Direct Connect, ExpressRoute)

1.3 Project Planning

- **Migration wave planning**
- RAID log
- Communication plan

2. AS/400 Discovery & Reverse Engineering



2.1 Source System Inventory

- **PF/LF file inventory**
- DDS extraction
- DB2/400 schema mapping

2.2 Code & Logic Analysis

- **RPG III/IV logic analysis**
- CL program mapping
- Business rule extraction

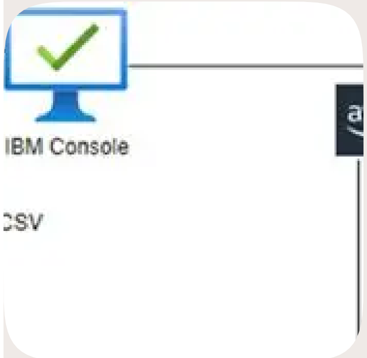
2.3 Operational Dependencies

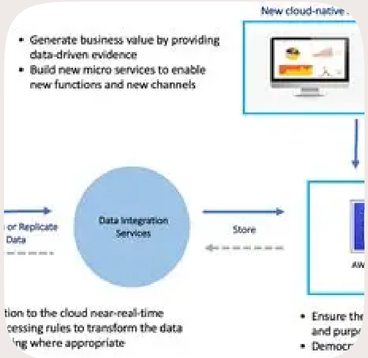
- Job scheduler mapping
- Batch cycle analysis
- Downstream consumer identification

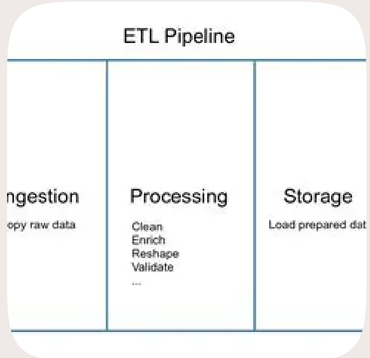
2.4 Data Profiling

- **Packed/zoned/EBCDIC profiling**
- Nullability & constraint analysis
- Data quality scoring

3. Extraction Framework (Full + Incremental)







3.1 Extraction Architecture

- DB2/400 extraction design
- File format standards (CSV, fixed-width, Parquet)
- Metadata framework

3.2 Full Load Extraction

- Parallel unload scripts
- EBCDIC → UTF-8 conversion
- Row-count & checksum generation

3.3 Incremental Extraction (CDC)

- Journaling-based CDC
- Timestamp-based deltas
- Change capture metadata

3.4 Automation

- Retry & checkpointing
- Logging & audit trails
- Scheduling

4. Cloud Landing Zone & Staging

4.1 Landing Zone Setup

- Raw zone

- Normalized zone
- Validated zone

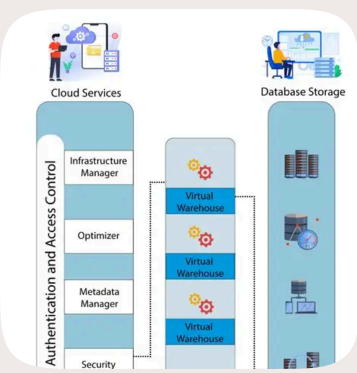
4.2 Storage & Access

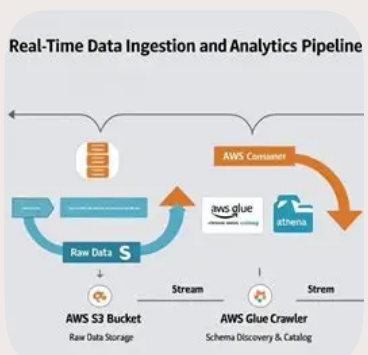
- S3/ADLS/GCS bucket creation
- Encryption policies
- IAM roles

4.3 Data Normalization

- Data type normalization
- Fixed-width parsing
- Encoding conversion

5. Snowflake Ingestion





5.1 Snowflake Stages

- External stage creation
- File format definitions
- Storage integration

5.2 Batch Ingestion

- COPY INTO pipelines
- Validation mode
- Error handling

5.3 Continuous Ingestion

- **Snowpipe auto-ingest**
- Event notifications
- Monitoring

5.4 Incremental Ingestion

- **Streams & Tasks**
- CDC merge logic
- Watermarking

6. Transformation (DBT + Snowflake ELT)

6.1 DBT Project Setup

- Repository structure
- Environment configuration
- CI/CD integration

6.2 Staging Models

- **DBT staging models**
- Column renaming
- Type casting

6.3 Intermediate Models

- **DBT intermediate models**
- Joins & business logic
- Surrogate key generation

6.4 Mart Models

- **DBT marts**
- Fact & dimension tables
- SCD Type 1/2

6.5 Optimization

- Clustering keys
- Micro-partition tuning
- Warehouse sizing

7. Validation & Reconciliation

7.1 Schema Validation

- **Schema checks**
- Column count & type matching

7.2 Metric Validation

- **Row counts, sums, distincts**
- Aggregation checks

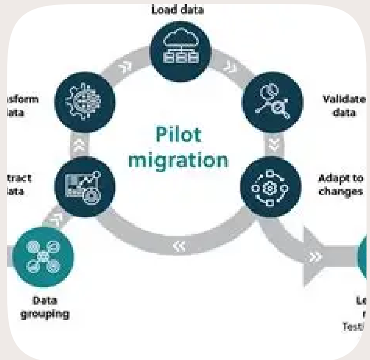
7.3 Row-Level Validation

- **Hash-based reconciliation**
- Exception reporting

7.4 Functional Validation

- **Legacy vs Snowflake output comparison**
- Business sign-off

8. Cutover & Go-Live



8.1 Cutover Planning

- Freeze window
- Final CDC replay
- Parallel run

8.2 Go-Live Execution

- Switch downstream systems
- Monitor ingestion & transformations
- Validate KPIs

8.3 Rollback Strategy

- Rollback criteria
- Legacy reactivation plan

9. Decommissioning & Optimization

9.1 Legacy Decommissioning

- PF/LF retirement
- RPG/CL job shutdown
- LPAR decommission

9.2 Snowflake Optimization

- Warehouse right-sizing
- Storage tiering

- RBAC/ABAC hardening

9.3 Documentation

- Final architecture
- Operational runbooks
- Knowledge transfer